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RV COLLEGE OF ENGINEERING
(Autonomous Institution Affiliated to VTU)
III Semester B.E. April -2024 Examinations
DEPARTMENT OF MATHEMATICS
STATISTICS, LAPLACE TRANSFORMS AND NUMERICAL METHODS
(Common to AS, BT, CH, IEM, ME)
(2022 SCHEME)

MODEL QUESTION PAPER

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART – A

1	1.1	A frequency distribution curve is said to be 'platykurtic' if ___________	01
	1.2	If two variables x and y are perfectly correlated then, coefficient of correlation ' r' is _________	01
	1.3	If the equations of regression lines of two variables x and y are $x - y = 0$, $4x - y = 3$ then the means of x and y are_____ and _____	02
	1.5	The real and imaginary parts of the function $f(z) = z + e^z$ is _____ and ________	02
	1.6	The residue of $f(z) = \frac{1}{(z+2)(z+3)}$ at the pole $z = 3$ is _________.	02
	1.7	Compute: $L[e^{3t}\cosh(2t)]$ is ____________.	02
	1.8	The Laplace transform of the signal $f(t) = t\,\delta(t-3)$ is _____.	02
	1.9	Evaluate: $L^{-1}\left[\frac{3}{s^2-4}\right]$	02
	1.10	If $F(s) = \frac{1}{s^2+2s+7}$, then $L^{-1}[F(s)] =$ ________.	02
	1.11	The forward difference approximation for u_t and central difference approximation u_x are _________ and ___________	02
	1.12	The 1D wave equation $c^2\,u_{xx} - u_{tt} = 0$ is classified as _______	02

PART – B

UNIT I

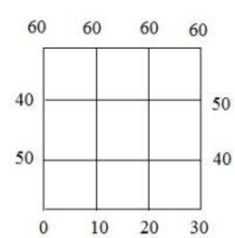
<table>
<tr><td>2</td><td>a</td><td>Compute the first four moments about the mean of the frequency distribution given below:
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td></tr>
</table></td><td>10</td></tr>
<tr><td></td><td>b</td><td>The following table gives ranking of 10 students according to their achievements in both the laboratory and lecture portions of an engineering course. Find the coefficient of rank correlation.
<table>
<tr><td>Lab(x)</td><td>8</td><td>3</td><td>9</td><td>2</td><td>7</td><td>10</td><td>4</td><td>6</td><td>1</td><td>5</td></tr>
<tr><td>Lecture(y)</td><td>9</td><td>5</td><td>10</td><td>1</td><td>8</td><td>7</td><td>3</td><td>4</td><td>2</td><td>6</td></tr>
</table></td><td>6</td></tr>
</table>

		UNIT - II	
3	a	Show that the function $f(z) = z^2$ is analytic and hence obtain its derivative.	8
	b	Expand the function $f(z) = \frac{z}{(z+1)(z+3)}$ as power series in the region $1 < \lvert z \rvert < 3$	8
		OR	
4	a	Construct an analytic function $f(z) = u + iv$, if the real part of the function is $u = x^2 + z + y - y^2$	8
	b	Evaluate $\int_c \frac{3z-4}{z(z-1)(z-2)} dz$ by using the residue theorem if $C: \lvert z \rvert = 3/2$	8

		UNIT - III	
5	a	For the periodic function $f(t) = \begin{cases} 1, & 0 < t < \frac{a}{2} \\ -1, & \frac{a}{2} < t < a \end{cases}$, where $f(t+a) = f(t)$, find $L[f(t)]$. Sketch the graph of the function.	8
	b	Find (i) $L[te^{-3t} + \sin(t)\sin(3t)]$ (ii) $L\left[\frac{\sin(2t)}{t}\right]$.	8
		OR	
6	a	Evaluate: $L\left[\left(2\sqrt{t} - \frac{1}{3\sqrt{t}}\right)^3 + t\cos(2t)\right]$	8
	b	Express the function in terms of unit step function and hence, obtain the Laplace transform of $f(t) = \begin{cases} \cos(t), & 0 < t < \pi \\ 1, & \pi < t \le 2\pi \\ \sin(t), & t \ge 2\pi \end{cases}$	8

		UNIT - IV	
7	a	Find the inverse Laplace transform of the following: (i) $\left[\frac{3s+2}{2s^2-4s+3}\right]$ (ii) $\left[\frac{2s^2-6s+5}{s^3-6s^2+11s-6}\right]$	8
	b	Employ convolution theorem to find $L^{-1}\left[\frac{1}{(s+1)\,s^2}\right]$.	8

		OR	
8	a	Find the inverse Laplace transform of the following: (i) $\log_e\left(\frac{s+a}{s+b}\right)$ (ii) $\frac{s\,e^{-\frac{s}{2}}+\pi\,e^{-s}}{s^2+\pi^2}$	8
	b	Using Laplace transform solve the problem of undamped forced vibrations of a spring governed by the equation $\frac{d^2x}{dt^2}+\beta^2x=A\sin\omega t\,, x(0)=x_0, x'(0)=v_0$.	8

		Unit - V	
9	a	Determine the values of $u(x,t)$ satisfying the parabolic equation $\frac{\partial u}{\partial t}=4\frac{\partial^2 u}{\partial x^2}$ and the boundary conditions $u(0,t)=u(8,t)=0$ and $u(x,0)=4x-\frac{x^2}{2}$ at the points $x=i: i=0,1,2,\dots,8$ and $j=0,1,2,\dots,5$.	8
	b	The temperature $u(x,y)$ for a steady state heat flow in a square plate bounded by $x=0, y=0,\ x=4,\ y=4$ satisfies Laplace equation $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$. Taking a square mesh of size unity and the boundary values given by $u=0$ along $x=0$ and $y=0$, $u=x^3$ along $y=4$, $u=16y$ along $x=4$. Compute two iterations using Gauss-Seidel iterative method.	8
		OR	
10	a	The transverse displacement u of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $\frac{\partial^2 u}{\partial t^2}=4\frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u=0$ at $x=0$, $t>0$ and $u=0$ at $x=4$, $t>0$ and initial conditions $u=x(4-x)$ and $\frac{\partial u}{\partial t}=0$ at $t=0$, $0\le x\le 4$. Solve the equation numerically for one half period of vibration, taking $h=1$ and $k=\frac{1}{2}$.	8
	b	Determine the values of $u(x,y)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ at the pivotal points of a square region, with boundary values as shown in the following figure.  Carry out three iterations.	8